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“Design and Implementation of a Meta-Learning Based Multi-Agent System for Deep Learning Model Synthesis and Architecture Mutation”
Towards AGI: LLM-Powered Multi-Agent AI for Recursive Model Synthesis and Mutative Architecture Evolution

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Outline

1. Abstract
2. Introduction
3. Literature Survey
4. Probable Methodology/Algorithm
5. Future Plans

Abstract

01 Framework

Meta-learning multi-agent system for automated deep learning model synthesis.

02 Collaboration

Planner, CodeGen, Training, Evaluator, Evolver, Memory, and Reflexion agents design, train, and evolve architectures.

03 Technology

Built using PyTorch, DEAP, and Streamlit for adaptive learning and visualization.

04 Applications

Efficient architecture exploration on MNIST and CIFAR-10 with minimal human intervention.

Introduction

- **Project Title:** Design and Implementation of a Meta-Learning Based Multi-Agent System for Deep Learning Model Synthesis and Architecture Mutation.
- **Context:** Automating neural network design is key to advancing AI, overcoming manual design limitations.
- **Motivation:** LLMs and meta-learning can enable autonomous architecture synthesis and mutation.
- **Goal:** Develop a multi-agent system with Planner, CodeGen, Training, Evaluator, Evolver, Memory, and Reflexion Agents for recursive model synthesis.
- **Relevance to AGI:** Iterative model evolution via agents enhances adaptability, a step toward AGI.
- **Scope:** Initial implementation and testing will utilize benchmark datasets (e.g., CIFAR-10, MNIST) for validating agent-driven synthesis.

SL. No	Title of the paper	Author	Year	Methodology	Advantages	Disadvantages
1	Reflexion	Shinn et al.	2023	Verbal Reinforcement Learning	91% pass rate on HumanEval	Limited to coding tasks
2	ReAct	Yao et al.	2022	Reasoning + Acting	Improved decision-making across tasks	Needs broader testing
3	Voyager	Wang et al.	2023	LLM-based Skill Acquisition	Outperforms prior methods in Minecraft	Domain-specific (Minecraft)
4	Evaluating LLMs on Code	Chen et al.	2021	Code Evaluation	28.8% problem-solving rate on HumanEval	Varies with problem complexity
5	AlphaCode	Li et al.	2022	Diverse Code Generation	Top 54.3% in Codeforces competition	High computational cost
6	DARTS	Liu et al.	2018	Differentiable NAS	Competitive, less computation (CIFAR-10)	Potential suboptimal convergence
7	ENAS	Pham et al.	2018	Parameter Sharing NAS	Strong performance, less compute (CIFAR-10)	Shared parameters limit diversity
8	NEAT	Stanley & Miikkulainen	2002	Neuroevolution	Outperforms fixed-topology in control tasks	Scalability concerns
9	MAML	Finn et al.	2017	Meta-Learning	State-of-the-art few-shot (Omniglot)	Computationally intensive
10	Matching Networks	Vinyals et al.	2016	One-Shot Learning	Strong one-shot performance (Omniglot)	Dependent on embedding quality